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Brief Presentation

on

Design a 4-bit Multiplier using Modified Gate Diffusion Input Technology for Low Power Applications

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OBJECTIVE

- Multipliers are important components in digital systems.
- Conventional multipliers consume high power.
- This project uses **MGDI technology** to reduce power consumption.
- The design improves **speed and efficiency** in VLSI circuits.

INTRODUCTION

- Multipliers are used in **Digital Signal Processing and Micro Processors**.
- Traditional CMOS circuits require more transistors.
- Power consumption is a major problem in digital circuits.
- MGDl technique helps reduce **power and delay**.

LITERATURE SURVEY

- In earlier designs, multipliers were implemented using CMOS technology, which resulted in higher power consumption and larger area.
- Modified GDI (MGDI) further improves efficiency by reducing power, delay, and area in multiplier circuits.
- The proposed 4-bit MGDI multiplier provides better performance and is suitable for low-power VLSI applications.

EXISTING MODEL

- Conventional multipliers use **CMOS technology**.
- Here, we used **6** (3 PMOS&3NMOS) transistors.
- Consumes more power during operation.
- Circuit design becomes complex.

LIMITATIONS OF EXISTING MODEL

- Existing CMOS multiplier circuits **consume high power during operation.**
- The design requires **6 number of transistors** to implement the circuit.
- The circuit produces **higher propagation delay**, which reduces speed.
- The overall design **occupies more chip area** in VLSI I implementation.

PROPOSED DESIGN

- The proposed system designs a **4-bit multiplier** using **Modified Gate Diffusion Input (MGDI)** technology.
- MGDI technique reduces the number of transistors from **6 (CMOS)** to **2 (MGDI)** in logic gates.
- The design helps to **reduce power consumption** in digital circuits.
- The proposed multiplier improves **speed and efficiency** in VLSI applications.

CMOS TO MGDI IMPLEMENTATION

- Multiplier circuit is first designed using **CMOS logic gates**.
- CMOS gates require **more number of Transistors (6T)**.
- CMOS gates are replaced with **MGDI logic gates (2T)**.
- MGDI reduces **Transistor count, Power consumption, and Delay**.
- This improves **Speed and Efficiency of the multiplier**.

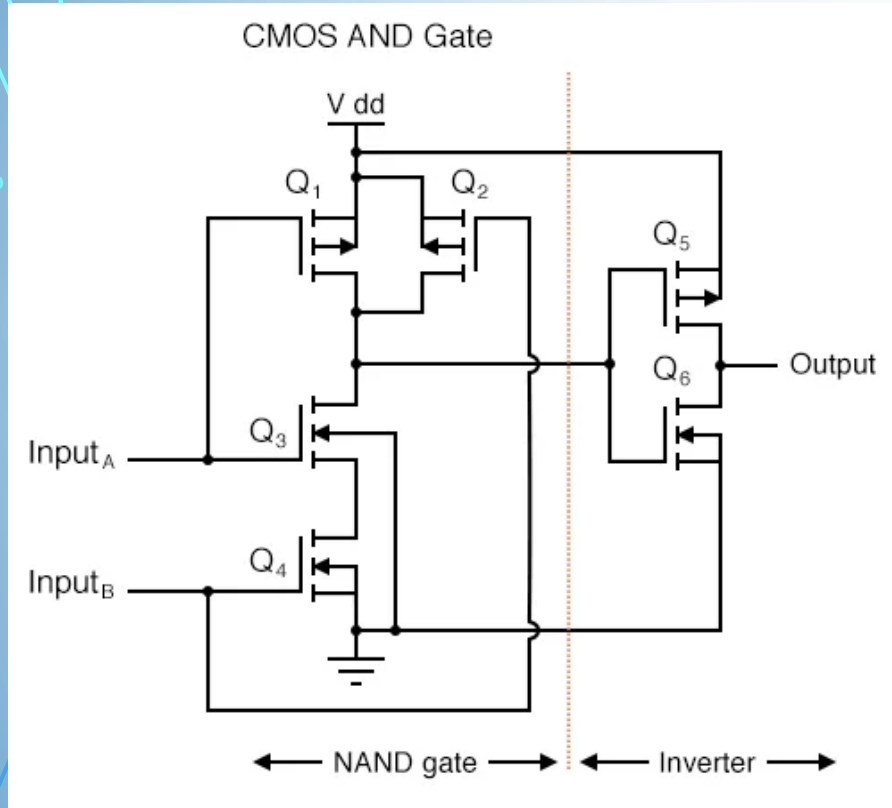


Fig (a): CMOS AND gate

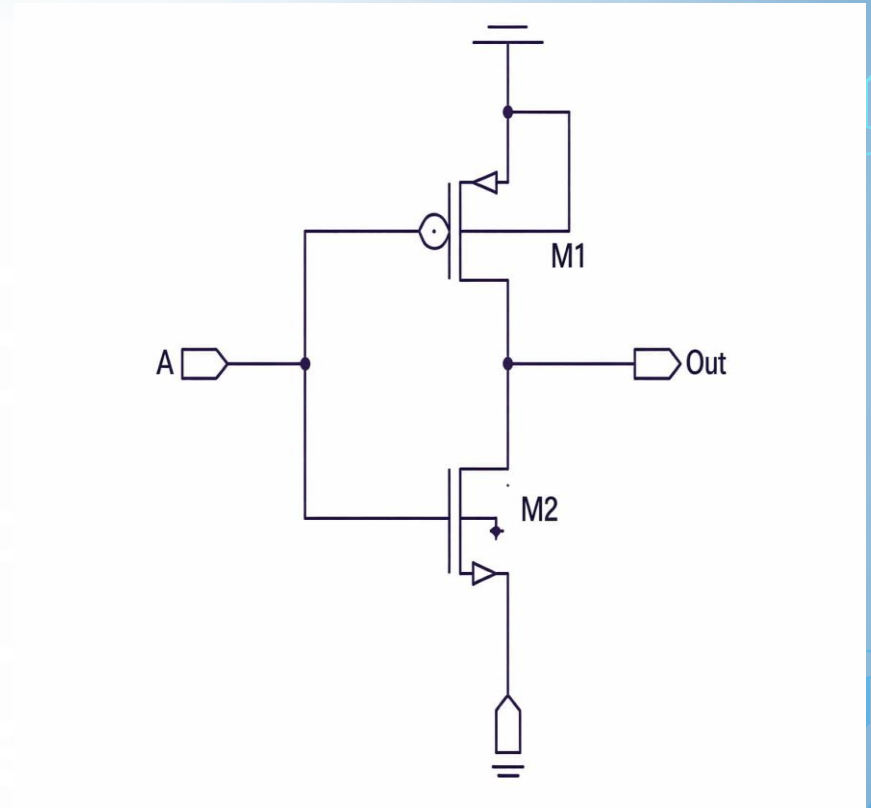


Fig (b): MGDI AND gate

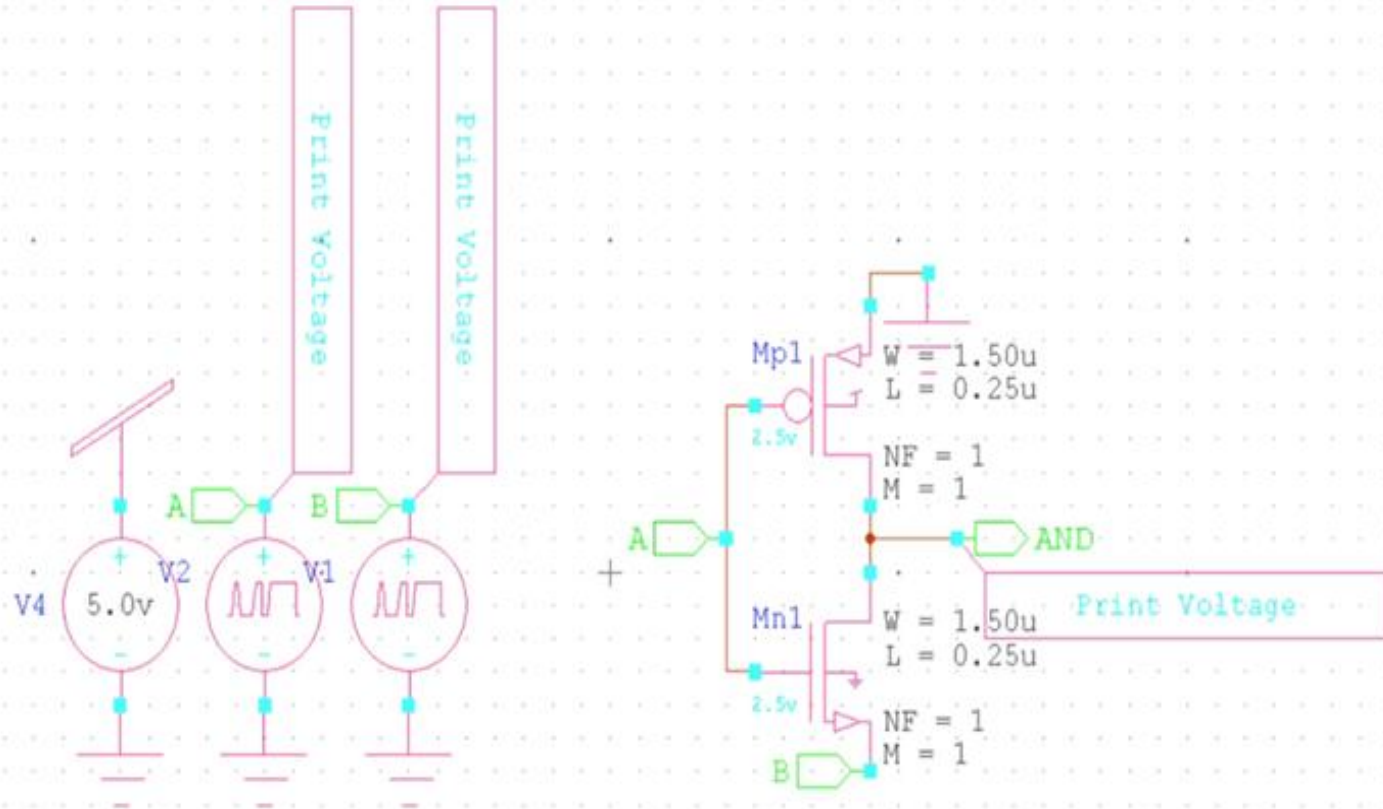


Fig : MGDI Schematic Diagram

WORKING MODEL

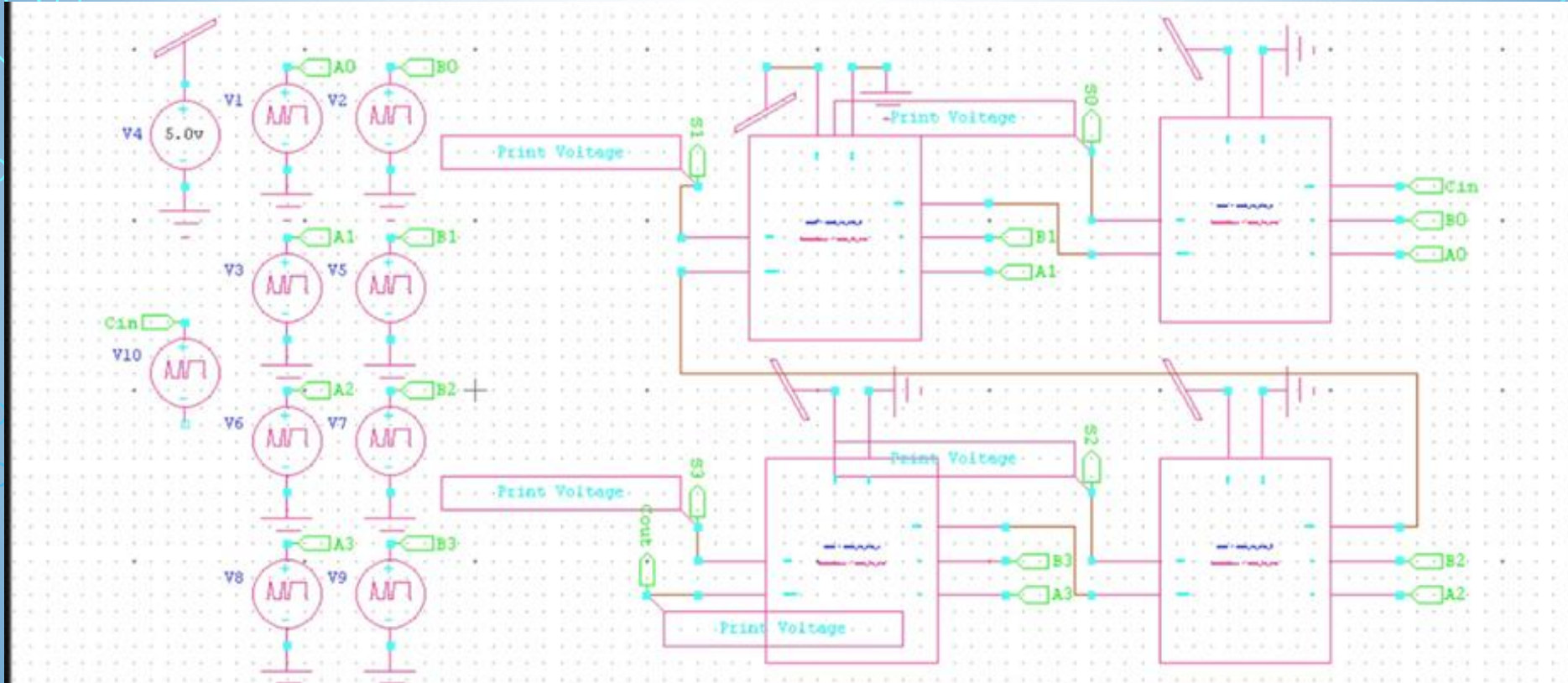


Fig : Multiplier Schematic Diagram

- **1. Input Stage:**

Two **4-bit binary numbers (A and B)** are given as inputs to the multiplier circuit.

- **2. Partial Product Generation:**

The **MGDI logic gates generate partial products** by multiplying each bit of the input numbers.

- **3. Addition Process:**

The generated partial products are **added using adder circuits** to perform the multiplication operation.

- **4. Final Output:**

The circuit produces the **8-bit multiplication result**, which represents the product of the two input numbers.

RESULT

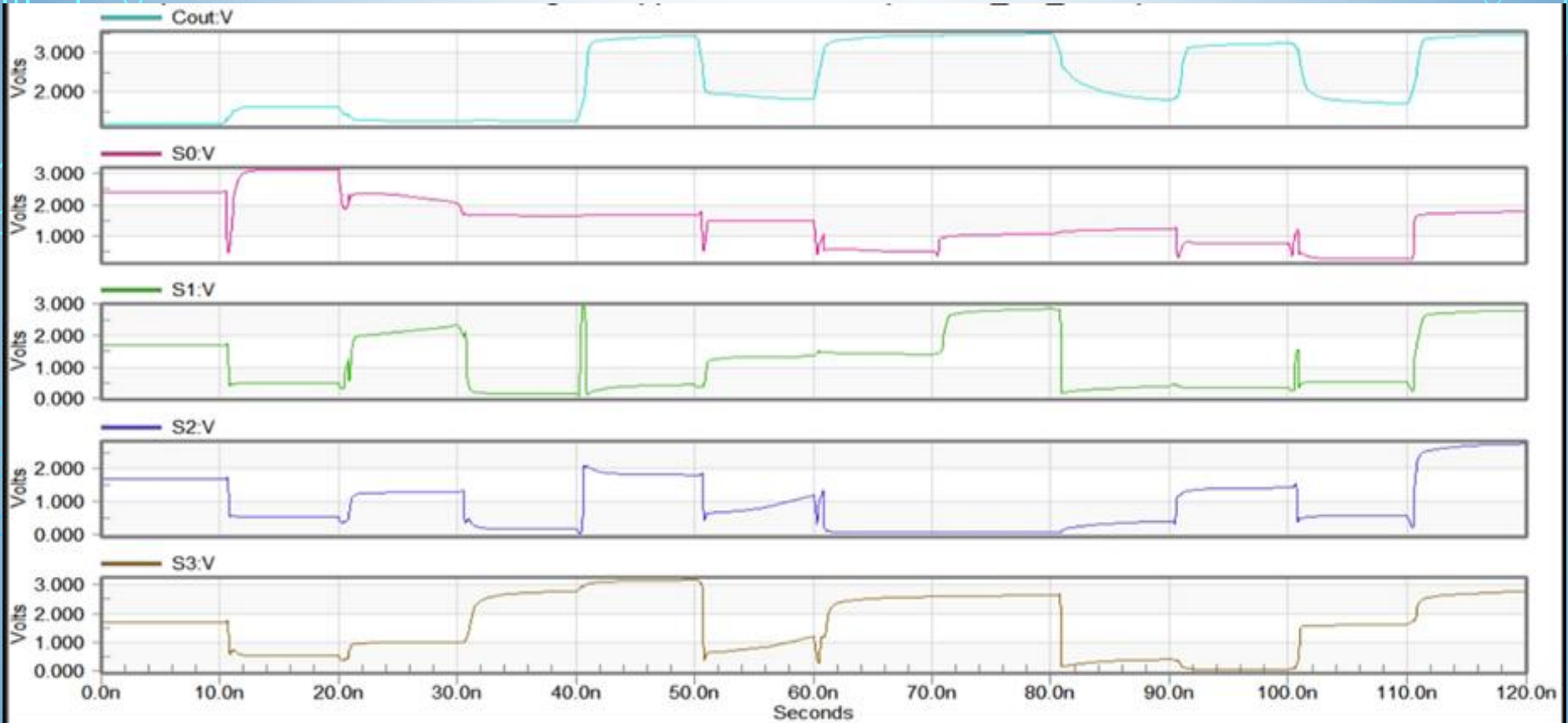


Fig: Final wave form

COMPARISION TABLE

PARAMETERS	EXISTING CMOS	PROPOSED MGDI
Transistor Count	6	2
Power Consumption	10 μ W	3 μ W
Propagation Delay	8 ns	3 ns
Area	30 μ m ²	10 μ m ²
Applications	MicroProcessors	Multipliers

CONCLUSION

- As we used MGDI Technology transistor count is reduced.
- Power consumption is reduced compared to CMOS design.
- Circuit performance and efficiency are improved.
- Suitable for low-power VLSI applications.

REFERENCES

- Bini Joy, N. Akshaya, and M. Sathia Priya, —Design of Multiplier Using GDI Technique, International Journal of Advanced Research in Science and Engineering, vol. 4, Special Issue 2, pp. 373–375, Feb. 2025.
- J. Aditya, N. Reddy, G. S. Krishna, and P. Bikki, “Novel High-Speed Vedic Multiplier”, Research Gate, Mar. 2024.
- Rama Krishna Reddy, R. Tarun, C. A. Kumar, and M. V. Rao, “An Efficient Adiabatic 4-Bit Array Multiplier for Low Power Applications,” International Journal of Advanced Research in Innovative Ideas in Engineering, vol. 10, no. 2, pp. 3711–3712, 2024.
- Reddaf et al., “Modeling of Schottky Diode and Optimal Matching Circuit Design for Low Power RF Energy Harvesting”, Heliyon, vol. 10, p. e27792, 2023.
- Vamshi Ponugoti, S. Oruganti, and S. Bopidi, “Design of Baugh Wooley Multiplier Using Full Swing GDI Technique,” EasyChair Preprint, no. 5374, Apr. 2023¹⁷



THANK YOU !